MATH 3603 Homework 4 Solutions

1. (a) Denote S as the state space and

$$I = \{(j_{k-1}, j_{k-2}, \cdots, j_0) \in S^k \mid j_{k-1} \neq i_{k-1}, \cdots, j_0 \neq i_0\},$$

and define $X_{[k-1]} := (X_{k-1}, \cdots, X_0)$. Since $\{X_n\}$ is a Markov chain, for all $a \in I$, we have

$$P(X_{n+1} = j \mid X_k = i, X_{[k-1]} = a) = P(X_{n+1} = j \mid X_k = i).$$

Then we have

$$\begin{aligned}
&P(X_{n+1} = j \mid X_k = i, X_{k-1} \neq i_{k-1}, \cdots, X_0 \neq i_0) \\
&= \frac{P(X_{n+1} = j, X_{k-1} \neq i_{k-1}, \cdots, X_0 \neq i_0 \mid X_k = i)}{P(X_{k-1} \neq i_{k-1}, \cdots, X_0 \neq i_0 \mid X_k = i)} \\
&= \frac{\sum_{a \in I} P(X_{n+1} = j, X_{[k-1]} = a \mid X_k = i)}{\sum_{a \in I} P(X_{[k-1]} = a \mid X_k = i)} \\
&= \frac{\sum_{a \in I} P(X_{n+1} = j \mid X_k = i, X_{[k-1]} = a) P(X_{[k-1]} = a \mid X_k = i)}{\sum_{a \in I} P(X_{[k-1]} = a \mid X_k = i)} \\
&= \frac{P(X_{n+1} = j | X_k = i) \sum_{a \in I} P(X_{[k-1]} = a \mid X_k = i)}{\sum_{a \in I} P(X_{[k-1]} = a \mid X_k = i)} \\
&= P(X_{n+1} = j \mid X_k = i) \\
&= P(X_{n-k} = j \mid X_0 = i).
\end{aligned}$$

(b) The equality does not hold in general. For example, consider any Markov chain with two classes: one being $\{0\}$ and the other being $\{1, 2, \ldots, \ell\}$, with $\ell \geq 2$ and $P_{ij} > 0$ for all $i, j \in \{1, 2, \ldots, \ell\}$. Take $n = 2$ and $k = 1$. Then clearly

$$P(X_2 = 1 \mid X_1 \neq 1, X_0 = 0) = 0,$$

while

$$P(X_1 = 1 \mid X_0 \neq 1) \neq 0.$$

2. Denote P as the transition probability matrix, and $p_0 = (\frac{1}{4}, \frac{1}{4}, \frac{1}{2})$ as the probability vector of initial distribution. Then the distribution of X_3 is given by $p_3 = p_0 P^3$, and $\mathbb{E}[X_3] = p_{3,2} + 2p_{3,3}$. In our case, we have

$$P^3 = \begin{pmatrix} \frac{91}{216} & \frac{23}{108} & \frac{79}{216} \\ \frac{14}{27} & \frac{5}{27} & \frac{8}{27} \\ \frac{23}{54} & \frac{7}{27} & \frac{17}{54} \end{pmatrix} = \begin{pmatrix} 0.4213 & 0.2130 & 0.3657 \\ 0.5185 & 0.1852 & 0.2963 \\ 0.4259 & 0.2593 & 0.3148 \end{pmatrix}.$$

Thus

$$p_3 = \begin{pmatrix} \frac{43}{96} & \frac{11}{48} & \frac{31}{96} \end{pmatrix} = \begin{pmatrix} 0.4479 & 0.2292 & 0.3229 \end{pmatrix},$$

and $\mathbb{E}[X_3] = \frac{7}{8} = 0.875$.

3. Let X_n denote the coin to be flipped after the n-th day of the initial flip. Clearly $\{X_n\}$ is a Markov chain with states $\{1, 2\}$. According to the rule of choosing the coin, the transition probability matrix of $\{X_n\}$ is given by

$$P = \begin{pmatrix} 0.7 & 0.3 \\ 0.6 & 0.4 \end{pmatrix}.$$

Denote $p_0 = (\frac{1}{2}, \frac{1}{2})$ as the initial distribution of X_0. Then the distribution of X_3 is given by

$$p_3 = p_0 P^3 = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} \frac{667}{1000} & \frac{333}{1000} \\ \frac{333}{500} & \frac{167}{500} \end{pmatrix} = \begin{pmatrix} \frac{1333}{2000} & \frac{667}{2000} \end{pmatrix} = \begin{pmatrix} 0.6665 & 0.3335 \end{pmatrix}.$$

Thus, the probability of flipping coin 1 in the third day after the initial flip is 0.6665.

4. For P_1 and P_2, one can check that all the states commutes. Since each of the Markov chain is finite, all the states are recurrent and there is only one class. For P_3, label the states as $\{1, 2, 3, 4, 5\}$. Clearly 4 only commutes with 5, 1 only commutes with 3, and 2 does not commutes with 1, thus there are three classes: $\{1, 3\}$, $\{4, 5\}$ and $\{2\}$. The states $1, 3, 4, 5$ are recurrent while 2 is transient.

5. Note that

$$X_{n+1} = \begin{cases} X_n, & \text{if } \xi_{n+1} \in \{\xi_1, \cdots, \xi_n\}, \\ X_n + 1, & \text{if } \xi_{n+1} \notin \{\xi_1, \cdots, \xi_n\}. \end{cases}$$

Thus

$$\begin{aligned}
&\mathbb{P}(X_{n+1}=j \mid X_n=i, X_{n-1}=i_{n-1}, \cdots, X_0=i_0) \\
&= \begin{cases} \mathbb{P}(\xi_{n+1} \in \{\xi_1, \cdots, \xi_n\}) & \text{if } j=i, \\ \mathbb{P}(\xi_{n+1} \notin \{\xi_1, \cdots, \xi_n\}) & \text{if } j=i+1, \\ 0 & \text{otherwise} \end{cases} \\
&= \begin{cases} \dfrac{i}{N} & \text{if } j=i, \\ 1-\dfrac{i}{N} & \text{if } j=i+1, \\ 0 & \text{otherwise.} \end{cases}
\end{aligned}$$

Note that the right hand side is independent from the values of $X_{n-1}, \ldots, X_0$. For each specific i and j, denote the right hand side above as P_{ij}. Thus

$$\begin{aligned}
&\mathbb{P}(X_{n+1}=j \mid X_n=i) \\
&= \sum_{i_0, \cdots, i_{n-1}} (\mathbb{P}(X_{n+1}=j \mid X_n=i, X_{n-1}=i_{n-1}, \cdots, X_0=i_0) \cdot \\
&\qquad \mathbb{P}(X_{n-1}=i_{n-1}, \cdots, X_0=i_0 \mid X_n=i)) \\
&= P_{ij} \sum_{i_0, \cdots, i_{n-1}} \mathbb{P}(X_{n-1}=i_{n-1}, \cdots, X_0=i_0 \mid X_n=i) \\
&= P_{ij}
\end{aligned}$$

Therefore, we conclude that $\{X_n\}$ is a Markov chain with transition probability

$$\mathbb{P}(X_{n+1}=j \mid X_n=i) = \begin{cases} \dfrac{i}{N} & \text{if } j=i, \\ 1-\dfrac{i}{N} & \text{if } j=i+1, \\ 0 & \text{otherwise.} \end{cases}$$

6. Since the Ehrenfest chain is finite (at most r states), it suffices to check its irreducibility. Note that for any $1 \leq \ell \leq r-1$ we have $P_{\ell,\ell+1} > 0$ and $P_{\ell,\ell-1} > 0$. Also we have $P_{0,1} > 0$ and $P_{r,r-1} > 0$. Thus for any states k, ℓ with $k < \ell$, we have

$$P_{k,\ell}^{\ell-k} \geq P_{k,k+1} \cdot \cdots \cdot P_{\ell-1,\ell} > 0.$$

And the same result holds clearly for $k > \ell$. Thus all the states in the Ehrenfest chain commutes, which completes the proof.